ZIQI TIAN

Email: ziqi.tian.gr@dartmouth.edu | Tel: (+1)603-573-5820

EDUCATION

Dartmouth College *Sept. 2025 - March 2027 (Expected)*
M.S. in Computer Science & Digital Art *Hanover, NH, USA*
Awarded 50% Tuition Scholarship

Sichuan University *Sept. 2021 - June 2025*
B.S. in Computer Science
GPA: 90/100, 3.8/4 *Chengdu, China*

RESEARCH EXPERIENCE

Visual Computing Lab, Dartmouth College Dec. 2025 - Present
Research Assistant, advised by Prof. Wojciech Jarosz *Hanover, NH, USA*

- Developed a framework based on GPIS that models the transition between surfaces and participating media to represent the micro-structure of snow&ice.
- Applied Fourier analysis to characterize the latent Gaussian autocorrelation structure of Bi-continuous media, improving color accuracy when reconstructing two-phase geometry compared to standard RTE model.
- Re-engineered a cubic-complexity model into a constant-time procedural noise system using sparse convolution and Taichi-based parallel computing, achieving roughly a 9000× speedup and making previously intractable simulations run at interactive rates.

PROJECT EXPERIENCE

Dartmouth Rendering Competition 2025 Sept. 2025 - Dec. 2025
Independent Developer *Hanover, NH, USA*

- Won **1st Place** in an individual developer competition sponsored by **NVIDIA**, building a physically-based Monte Carlo ray tracer from scratch.
- Implemented spectral rendering as the main feature, providing color-accurate results and ability to incorporate different color space into ray-tracing process.
- Built a Multiple Importance Sampling (MIS) and Next-Event Estimation (NEE) framework for variance reduction, and used BVH acceleration structures to speed up spatial queries in complex scenes.

WORK EXPERIENCE

Lilith Games Feb. 2025 - Apr. 2025
Technical Artist Intern *Shanghai, China*

- Built Python automation tools to batch-process and bind large visual data assets, streamlining the studio's research-to-production workflow.
- Developed real-time simulation systems that use stochastic modeling to represent heterogeneous environments in 3D engines.

Ubisoft Apr. 2024 - July 2024
Technical Animator Intern *Chengdu, China*

- Implemented a motion-matching system that searches a large database of motion-capture clips and blends the best match to the character's current movement and the player's input, producing smoother real-time animation; the system was later adopted for use in *Assassin's Creed: Shadows*.

TEACHING EXPERIENCE

Graduate Teaching Assistant, Dartmouth College *2026 - Present*

- **CS70 Foundations of Applied Computer Science**
- **CS129.06 Digital Tangible User Interfaces**

RESEARCH INTERESTS

My research focuses on deriving rendering algorithms to represent and simulate complex visual appearance. I am interested in understanding why things look the way they do, and how to design machine learning techniques that best capture the underlying optical features.

I'm also keen on building small hardware devices and understanding the algorithms hidden behind them. I enjoy working with any inventor who comes up with weird gadgets.

RENDERING WORK

Entangled — 1st Place, Dartmouth Rendering Competition 2025